

Mavolutide Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-987/C

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Deviations observed during the campaign were investigated and closed with no impact to product quality. No changes to the approved analytical procedures are proposed in this submission.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. No changes to the approved analytical procedures are proposed in this submission. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality. Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process

parameters identified during development.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. No changes to the approved analytical procedures are proposed in this submission.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. No changes to the approved analytical procedures are proposed in this submission. Cleaning verification swab results were below the health-based exposure limits derived for the product. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Deviations observed during the campaign were investigated and closed with no impact to product quality. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Deviations observed during the campaign were investigated and closed with no impact to product quality. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. The control strategy links critical quality attributes to critical process parameters identified during development. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The control strategy links critical quality attributes to critical process parameters identified during development.

P.4 Control of Excipients

For the commercial formulation F-987/C, Methocel E5 Premium LV (Hypromellose, release-controlling polymer) is used.

Hypromellose complies with the requirements of ICH Q8(R2).

Excipient Grade Selection

Methocel E5 Premium LV was selected as the commercial grade of hypromellose on the basis of compatibility with the active ingredient demonstrated in binary stress studies.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-8109 Rev 07) lists Hypromellose as Methocel K4M Premium CR.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Deviations observed during the campaign were investigated and closed with no impact to product quality.